RESEARCH PORTFOLIO

A.K.M. AMANULLAH

Turning uncertain data into defensible energy-system decisions

MECHANICAL ENGINEER / UNIVERSITY LECTURER / EARLY-CAREER RESEARCHER

Research emphasis: waste-to-hydrogen, process simulation, techno-economic and environmental analysis, thermoelectric cooling, anaerobic digestion, and water–energy systems.

01	04	02
published journal article	ICMIME 2026 accepted papers	additional research manuscripts

TARGET

Fully funded PhD in Mechanical Engineering — Fall 2027

SELECTED CREDENTIALS

- B.Sc. Mechanical Engineering, RUET (3.51/4.00)
- Seventh-semester GPA 3.86/4.00; 3rd of 181
- Lecturer, Textile Engineering, BUBT
- IELTS Academic 7.0 • October 2026

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Prepared November 2026. Conference papers marked accepted are forthcoming, not published; journal manuscript statuses are stated separately.

01 / RESEARCH IDENTITY

Research approach

The through-line: physical constraints + predictive models + uncertainty + actionable engineering trade-offs.

QUESTIONS I WANT TO ANSWER

- What happens to an optimum when inputs, operating conditions or prices depart from their nominal values?
- How can first-principles thermodynamics and data-driven models reinforce rather than contradict one another?
- Which measurements are necessary to turn a promising simulation into evidence of real-world performance?

METHODS APPLIED ACROSS PROJECTS

Prediction

ARIMA–XGBoost forecasting; regression and simulation-based response surfaces.

Physical modeling

Aspen Plus process flowsheets; energy and exergy reasoning; heat-transfer and mass-transfer fundamentals.

Uncertainty and optimization

Monte Carlo propagation; Sobol sensitivity analysis; Box–Behnken RSM; multi-objective and Pareto methods.

Decision framing

Levelized cost, economic risk, environmental boundaries, water and energy trade-offs.

SCOPE OF EVIDENCE

I distinguish published findings, peer-reviewed conference acceptances, draft manuscripts, simulation outcomes, and measurements. Numerical forecasts and response-surface predictions are not presented as plant-level validation; studies with unreplicated data are treated as exploratory.

RESEARCH ARC

Power-generation training → undergraduate energy research → integrated waste-to-hydrogen modeling → thermal-system optimization → multi-disciplinary sustainability and teaching.

02 / FLAGSHIP PUBLICATION

Waste to hydrogen, with uncertainty quantified

Published first-author research • International Journal of Hydrogen Energy, 2026

22 years	10,000	$3.80/kg
municipal-waste data	Monte Carlo draws	modeled LCOH

STUDY DESIGN

The research connects municipal solid waste projections in six Bangladeshi city corporations to a modeled anaerobic digestion–steam methane reforming (AD–SMR) pathway, a comparator landfill-gas pathway, and cost/carbon assessments. Forecasts, process models and scenarios are treated as linked layers rather than interchangeable sources of certainty.



Simplified analytical sequence reproduced from the published paper; schematic is newly drawn for this portfolio.

REPORTED, MODEL-BASED FINDINGS

- 26.2 Mt potential hydrogen production under the studied AD–SMR scenario (95% interval 21.4–31.7 Mt).
- Published study reports an estimated levelized hydrogen cost of $3.80/kg and cradle-to-gate carbon footprint of 2.1 kg CO_2-eq/kg H_2.
- Global sensitivity analysis identifies moisture and reformer efficiency as important economic drivers.

RESEARCH JUDGMENT / NEXT STEP

The next research challenge is validation and operational robustness: reconcile feedstock variability with process kinetics and equipment constraints, evaluate data/model transferability, and benchmark estimates against reliable plant or pilot measurements.

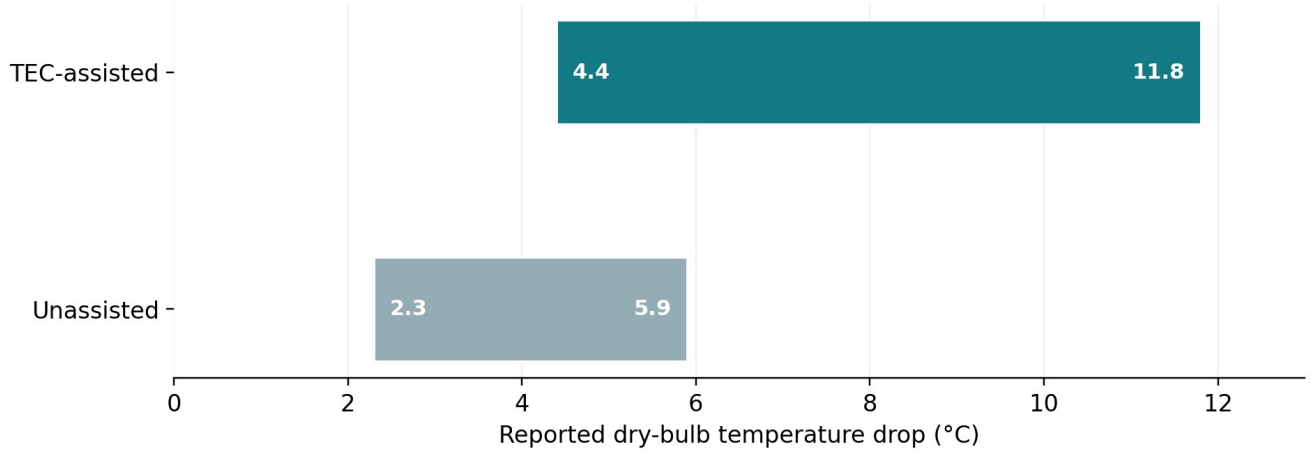
A.K.M. Amanullah and Md. Rabiul Islam Sarker, Int. J. Hydrogen Energy 254 (2026), 156339. DOI: 10.1016/j.ijhydene.2026.156339

03 / THERMAL ENGINEERING

Thermoelectric-assisted evaporative cooling

Journal manuscript in preparation • With Md. Emdadul Hoque and collaborators

Research objective: improve dry-bulb cooling under humid tropical conditions through thermoelectric precooling of recirculating pad water, while examining the associated electricity and heat-rejection penalties.



Ranges across 12 source-reported prototype comparisons; bars display minima and maxima, not confidence intervals. Average paired benefit reported: +2.63 °C.

04	29	2.63 °C
TEC modules	designed simulation runs	average paired gain

WHAT THE ANALYSIS DEMONSTRATES

- Four operating factors: TEC voltage, fan speed, evaporative-pad water flow and hot-side water flow.
- Quadratic response-surface models reconstructed from a 29-run Box–Behnken simulation design (reported R^2 0.9904–0.9997).
- Multi-objective compromises depend on how electricity-derived cost and emission objectives are weighted and whether criteria are double-counted.

BOUNDARY OF THE CLAIM

Excellent fit to simulation does not prove physical cooling capacity or room-scale performance. The manuscript documents heat-inventory inconsistencies and specifies additional humidity, air-flow and heat-flux measurements needed for independent validation.

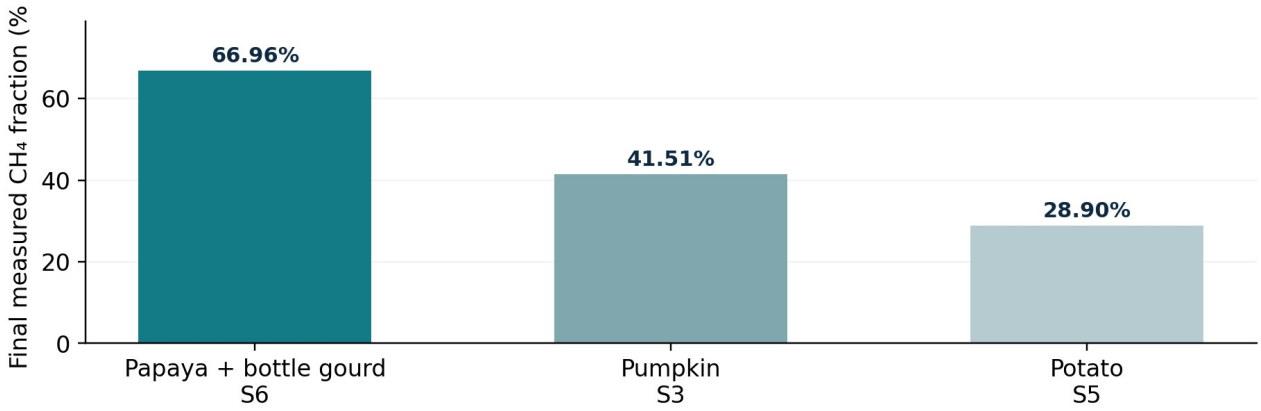
04 / WASTE AND WATER SYSTEMS

Energy–water research extensions

Two complementary studies connect operating conditions to conversion or removal performance.

ANAEROBIC CO-DIGESTION / ICMIME 2026 PAPER 318 / ACCEPTED

Eight cafeteria-vegetable-waste configurations were tracked for methane fraction and pH at 37 ± 1 °C over differing durations (20–130 days). Selected final values at day 130 show substantially different methane concentrations among setups.



Selected 130-day endpoints only; methane fraction is NOT methane volume or yield. No replicates or total gas volume were recorded.

FLOW-ELECTRODE CAPACITIVE DEIONIZATION / ICMIE 2026 MANUSCRIPT / ACCEPTANCE UNCONFIRMED

Model → parametric response → Pareto frontier MODELING STUDY

Semi-empirical relationships connect voltage, flow rate and carbon loading with predicted ion-removal efficiency and energy use. The manuscript reports model conditions with >90% predicted removal around 1.1–1.3 V, with sharply rising energy needs at higher voltage. These are modeling claims, not new laboratory experiments.

PHD SIGNIFICANCE

Both studies motivate research that measures what the model actually needs: gas volume and substrate chemistry for digestion; experimental validation, energy metering and electrode stability for FCDI. They reinforce a cross-cutting interest in systematic uncertainty and model–experiment agreement.

05 / CONFERENCE CONTRIBUTIONS

Four accepted ICMIME 2026 papers

Rajshahi University of Engineering & Technology • 9–11 December 2026 • proceedings forthcoming

318 / Anaerobic co-digestion and methane/pH dynamics WASTE-TO-ENERGY

Second author and corresponding author. Long-duration observations across eight cafeteria-waste configurations; careful distinction between methane concentration and methane yield.

331 / Conventional versus ozone-assisted denim washing RESOURCE EFFICIENCY

Second author. Recipe water requirement decreased from 360 L to 205 L (43.1%) for each ozone-based route, alongside fabric-strength trade-offs; not a general claim of superior performance.



Reported process-recipe water consumption, not a full-factory water footprint.

338 / Alternative reactive-dyed cotton stripping WASTEWATER

Second author and co-corresponding author. Alternative NaCl–NaOH samples recorded COD 176 mg/L versus 205 mg/L for conventional stripping; the dataset does not establish salinity or toxicity superiority.

339 / Bio-based multifunctional cotton finish BIO-BASED MATERIALS

Second author. External laboratory reported 99.27% reduction against S. aureus under a specified test; inconsistent flame-spread records preclude verified flame-retardancy claims.

Acceptance statements are based on the applicant's November 2026 confirmation. No paper above is described as already published, indexed, or assigned a DOI.

06 / TEACHING AND DIRECTION

A researcher who teaches and learns

The goal is a PhD built around testable hypotheses, credible models and measured performance.

ACADEMIC EXPERIENCE

Lecturer — BUBT, Department of Textile Engineering | 2026–present

Teaching includes fundamentals of mechanical engineering, machine dynamics and design, automation and robotics in textiles, and engineering drawing laboratory. Sustainability engagement includes membership in BUBT SDG Team 2 (SDG 4; linked SDGs 11 and 12).

Mechanical Engineering Instructor — BDIT | 2024–2026

Delivered courses, laboratory activities, assessments and student support. Industrial training at Ashuganj Power Station (March–April 2023) strengthened exposure to power generation operations.

PROPOSED DOCTORAL TRAJECTORY

Research question

How do feedstock and operating uncertainty shift the design and economic performance of low-carbon conversion pathways?

Method development

Combine validated thermodynamic/process models with constrained learning, Bayesian or stochastic uncertainty analysis, and robust optimization.

Validation plan

Use lab/pilot data or independently documented plant observations to test mass and energy balances, predictive error and transferability.

Broader impact

Publish reproducible methods and train engineering students in model-based decisions for energy and environmental systems.

SELECTED RESEARCH RECORD

Published: Amanullah & Sarker, International Journal of Hydrogen Energy (2026), article 156339. Publisher DOI

Forthcoming conference: ICMIME 2026 papers 318, 331, 338 and 339; acceptance reported; not yet proceedings publications. Official ICMIME site

In preparation / status to confirm: Thermoelectric-assisted cooler journal manuscript; FCDI/ICMIE conference manuscript.

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